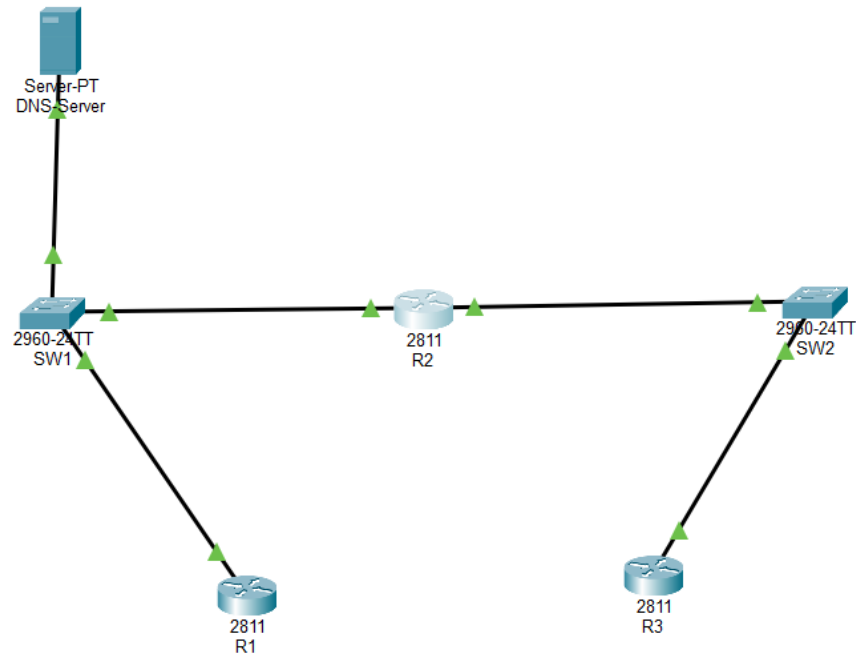


Week 3: Life of a Packet (10 Marks)



16. R1:

```
Router#ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip domain-lookup
Router(config)#ip name-server 10.10.10.10
Router(config)#ip domain-name lab.local
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

R2:

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip domain-lookup
Router(config)#ip name-server 10.10.10.10
Router(config)#ip domain-name lab.local
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

R3:

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip domain-lookup
Router(config)#ip name-server 10.10.10.10
Router(config)#ip domain-name lab.local
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

17.

```
Router#ena
Router#ping 10.10.20.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.20.1, timeout is 2 seconds:
..!!!
Success rate is 60 percent (3/5), round-trip min/avg/max = 0/1/2 ms

Router#
```

18.

```
Router#clear arp-cache
Router#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 10.10.10.1 - 0090.2B0B.A701 ARPA FastEthernet0/0
Internet 10.10.10.2 0 0090.213C.9201 ARPA FastEthernet0/0
Internet 10.10.10.10 0 0005.5E28.B6E5 ARPA FastEthernet0/0
Router#
Router#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/3/16 ms

Router#
```

19. R1:

```
Router#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 10.10.10.1 - 0090.2B0B.A701 ARPA FastEthernet0/0
Internet 10.10.10.2 4 0090.213C.9201 ARPA FastEthernet0/0
Internet 10.10.10.10 4 0005.5E28.B6E5 ARPA FastEthernet0/0
Router#
```

R2:

```
Router>ena
Router#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 10.10.10.1 13 0090.2B0B.A701 ARPA FastEthernet0/0
Internet 10.10.10.2 - 0090.213C.9201 ARPA FastEthernet0/0
Internet 10.10.20.1 13 0006.2A10.1501 ARPA FastEthernet0/1
Internet 10.10.20.2 - 0090.213C.9202 ARPA FastEthernet0/1
Router#S
```

R3:

```
Router>ena
Router#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 10.10.20.1 - 0006.2A10.1501 ARPA FastEthernet0/0
Internet 10.10.20.2 14 0090.213C.9202 ARPA FastEthernet0/0
Router#
```

20. ARP table shows only the next hop IP address and which interface to send the packet out. An ARP entry gives you the MAC address of that next hop.

21. Because IP routing and Ethernet delivery are 2 different jobs